

PROBLEM SET – IV

Session:	2024-25(Even Sem.)	Semester:	II
Course:	B.Tech.	Subject Code:	UMA2003
Branch:	CS, CE,ME,EC,EE	Subject Name:	Mathematics - II
Group:	All Groups	Name of Faculty members:	Dr. V. N. Pathak Dr. Prachi Srivastava Dr. Anurag Agnihotri Dr. Mukesh Awasthi Dr. Akhilesh Kumar Mishra Dr. Anil Kumar Dr. Awanish Kumar Dr. Puneet Kumar Gupta Dr. Pooja Gautam
Unit:	IV	Topic Covered:	O.D.E. of higher orders

1. Solve $(D^4 - 81)y = 0$ (BL3)
2. Solve $(D^4 + D^2 + 1)y = 0$ (BL3)
3. Solve $(D^2 + D + 1)^2 y = 0$ (BL3)
4. Solve $(D^2 + 4)y = \tan 2x$ (BL3)
5. Solve $(D^2 + 1)y = \sec x$ (BL3)
6. Solve $(D^2 + 6D + 9)y = 2e^{-3x}$ (BL3)
7. Solve $(D^2 - 1)y = \cosh x$ (BL3)
8. Solve $(D^2 - 4)y = \cos^2 x$ (BL3)
9. Solve $(D^2 + 9)y = \cos 2x + \sin 2x$ (BL3)
10. Solve $(D^3 + 3D^2 + 2D)y = x^2$ (BL3)
11. Solve $(D^4 + 8D^2 + 16)y = 16x + 256$ (BL3)
12. Solve $(D^2 - 5D + 6)y = e^{4x}(x^2 + 9)$ (BL3)
13. Solve $(D^2 + 1)y = x^2 \sin 2x$ (BL3)
14. Solve $(D^2 - 4)y = e^x \sin 2x$ (BL3)
15. Solve $(x^2 D^2 - xD + 3)y = x^2 \log x$ (BL3)
16. Solve $(x^2 D^2 - xD + 4)y = \cos(\log x) + x \sin(\log x)$ (BL3)
17. Solve $(x^3 D^3 + 3x^2 D^2 + xD + 1)y = x \log x$ (BL3)
18. Solve $(D^2 - 4D + 4)y = 8x^2 e^{2x} \sin 2x$ (BL3)
19. Apply method of variation of parameters to solve (BL3)

$$(i) y'' + n^2 y = \sec nx$$

$$(ii) y'' - 4y' + 3y = \frac{e^x}{1 + e^x}$$

$$(iii) y'' + 2y' + y = e^{-x} \log x$$

Supplementary Question:

20. Apply method of variation of parameters to solve (BL3)

$$x^2 y'' + 4xy' + 2y = e^x$$

Bloom's Taxonomy:

BL1: Remembering **BL2:** Understanding **BL3:** Applying **BL4:** Analyzing **BL5:** Evaluating

Text Books

1. E. Kreyszig, “Advanced Engineering Mathematics”, 9th Edition, John Wiley and Sons, New York, 2006.
2. R.K. Jain and S.R.K. Iyengar, “Advanced Engineering Mathematics”, Narosa Publishing House, 2008.

Reference Books

1. Peter V. O’ Neil,” Advanced Engineering Mathematics”, Thomson (Cengage Learning), 2007.
2. Veerarajan T., Engineering Mathematics for first year, Tata McGraw Hill, New Delhi, 2008.

(Signature of the Faculty Member with Date)

(Signature of the Dean with Date)